

**DEPARTMENT OF CHEMICAL & BIOMOLECULAR ENGINEERING
NORTH CAROLINA STATE UNIVERSITY**

CHE 330

Homework Set 6

Note: It is not required to show all calculations for means and standard deviations in this homework set. However, points will be deducted if these values are incorrect.

Problem 1 (20%) A fiber researcher tested the strength of fibers containing varying content of synthetic materials and obtained the data provided in the table below. **(a, 15%)** Perform an ANOVA on the data set to determine if the fiber strength depends on the content of synthetic material at a 0.01 significance level. **(b, 5%)** Confirm your results with the corresponding results provided by Excel®'s ANOVA (Single Factor) tool and include a printout of the results along with your hand solution to Part a). Be sure to specify the appropriate value of significance level over the default value of 0.05 in the Excel® tool.

Table 1. Fiber strength (kpsi) for varying synthetic material content.

Sample	% Synthetic Materials			
	15%	20%	25%	35%
1	8.7	14.5	16.0	9.0
2	9.8	17.3	13.8	14.3
3	12.2	14.9	18.3	10.0
4	12.9	15.5	17.2	14.3
5	9.5	12.7	18.0	7.1

Problem 2 (20%) A catalyst researcher studied the performance of hydrogenation catalysts with four different promoters and obtained the data provided in the table below. **(a, 15%)** Perform an ANOVA on the data set to determine if catalyst performance is dependent on the promoter types at a 0.05 significance level. **(b, 5%)** Confirm your results with the corresponding results provided by Excel®'s ANOVA (Single Factor) tool and include a printout of the results along with your hand solution to Part a).

Table 2. CO conversion by catalysts containing different promoters.

Trial	Fe Promoter	Mn Promoter	Pt Promoter	Pd Promoter
1	60.80	68.70	102.60	87.90
2	57.00	67.70	102.10	84.20
3	65.00	74.00	100.20	83.10
4	58.60	66.30	96.50	85.70
5	61.70	69.80	98.80	90.30

Problem 3 (35%) A kettle filled with boiling water is placed on a counter to cool. The room temperature is 20°C. The combined mass of the water and kettle is 2.0 kg, and their heat capacity is 1.25 kcal/(kg°C). The rate at which heat is lost from the kettle is given by Q (kcal/min) = $U \cdot A \cdot (T - T_o)$ where U (kcal/(m²min)) is the heat transfer coefficient, A is the heat

transfer area (m^2), and T and T_0 are, respectively, the kettle and room temperatures ($^{\circ}\text{C}$). Analysis suggests that T should vary with the cooling time t (min) as described by

$$T = [20 + 80 \cdot \exp(-0.40 \cdot U \cdot A \cdot t)]^{\circ}\text{C} \quad (1)$$

Measurements give the data shown in Table 3.

Table 3. Kettle temperature as a function of elapsed time.

t, min	0	3.0	7.5	15.0	30.0
T, $^{\circ}\text{C}$	100.0	87.0	74.3	53.3	35.8

(a, 20%) Complete a linearization transform of the non-linear expression given in Equation 1 to determine UA from the data. Plot the data and the regression results on a single figure (you can use Excel for this). **(b, 15%)** Determine the 95% confidence interval for UA . With the exception of your plot produced in Part a, be sure to *show your work* for Parts a and b, i.e. do not use Excel or an iterative method to solve.

Problem 4 (25%) A homogeneous chemical reaction $A \rightarrow B$ is known to follow the kinetic relationship

$$-r_A = kC_A^n \quad (2)$$

where r_A is the rate of reaction, k is the rate constant, C_A is the concentration of Species A, and n is the exponential factor. Testing provides kinetic data shown in Table 4.

Table 4. Kinetic data for a homogeneous reaction.

C_A (mol/L)	4	2.25	1.45	1.0	0.65	0.25	0.006
$-r_A$ (mol/Ls)	0.398	0.298	0.238	0.198	0.158	0.098	0.048

(a, 5%) What type of function is described by Equation 2? **(b, 20%)** Use a linearization transform of the non-linear expression given in Equation 2 to determine the values of n and k . Be sure to *show your work* for Part b, i.e. do not use Excel or an iterative method to solve.